

Architecture Selection Justification

Patient Health Record Consent Management System — Lab 3

Architecture Selection: We chose Layered Architecture for the Patient Health Record Consent Management System.

Architectural Choice: The system is organized into Presentation, Business/Application, and Data & Security layers. The Business/Application layer handles consent management, access control/verification, and expiry/notification responsibilities. The Data & Security layer contains consent data, protected patient records, and the append-only audit trail.

Reason 1 — Separation of Concerns: Consent granting/revocation, access validation, notifications, and data storage have distinct responsibilities. Separating these concerns makes the design easier to understand and isolates changes between layers.

Reason 2 — Controlled Access: Patient health records require controlled access. A dedicated application layer can validate consent status and doctor verification before record access, while the data/security layer protects stored records and audit information.

Security Advantage: The architecture gives clear locations for enforcing TLS 1.3 for data in transit and AES-256 for patient health records at rest. The append-only audit trail is also represented as a distinct component for accountability.

Performance Benefit: Separating presentation from business and data operations avoids placing all responsibilities in one component. Expiry/notification processing can operate separately from the user-facing interface, while access validation can occur without directly exposing the underlying record store.

Basis: The Lab 3 handout asks students to compare Layered, Microservices, and Client-Server styles and select an architecture based on the assigned scenario. This selection is based on that guidance and the finalized project requirements. No programming language, database, cloud provider, or deployment technology has been assumed.